Department of Computer Science and Engineering, IIT Delhi

COL334 Assignment 4

Make the Network Find its Way

Static Routing, RIPv2 and OSPF in Cisco Packet Tracer

Name	**Entry number**
Yashwardhan Singh Choudhary	2024CS10300
Tejasvi Shrivastava	2024CS10582

Packet Tracer version: 9.0.1.0858

Setup

The network (four Cisco 2911 routers R1–R4, four PCs, five router-to-router links and four LANs) was built exactly as in Tables 1 and 2 of the assignment and saved as `base.pkt`. R2 is the DCE end of the R2–R3 serial link (clock rate 128000), Serial0/0/1 on R2 and R3 is shut down, and every router carries the banner `2024CS10300 2024CS10582`. Before saving, every router pinged the far end of each of its links and every PC pinged its default gateway. Parts A, B and C each start from a fresh copy of `base.pkt`. In all route-table screenshots the route to LAN D (192.168.4.0/24) is marked with a red box.

1 Part A: Static Routing

Each router has exactly three static routes in next-hop form, one to each remote LAN. The next hop is the neighbour that owns the LAN, except R1→LAN D and R4→LAN A, which go via R2.

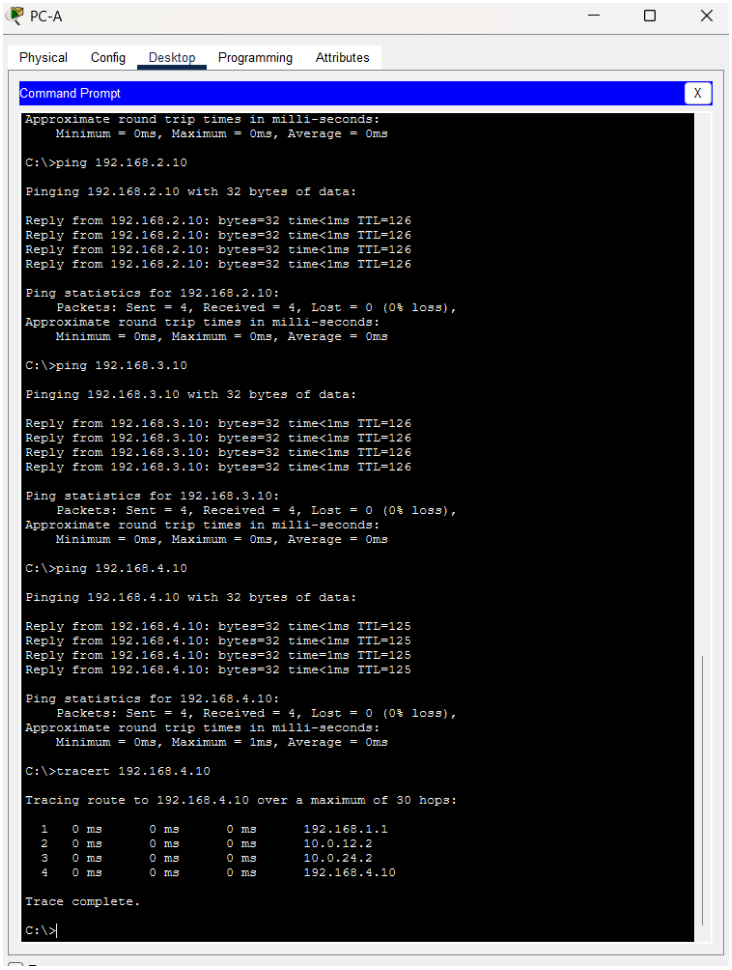


Figure 1: Part A: PC-A pings PC-B (192.168.2.10), PC-C (192.168.3.10) and PC-D (192.168.4.10) successfully, followed by `tracert 192.168.4.10`.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.13.0/30 is directly connected, GigabitEthernet0/2
L       10.0.13.1/32 is directly connected, GigabitEthernet0/2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S    192.168.2.0/24 [1/0] via 10.0.12.2
S    192.168.3.0/24 [1/0] via 10.0.13.2
S    192.168.4.0/24 [1/0] via 10.0.12.2

R1#
```

Figure 2: Part A: R1 `show ip route`. The static route to LAN D (boxed) is `S 192.168.4.0/24 [1/0] via 10.0.12.2`.

Q1. Addresses printed by `tracert 192.168.4.10` on PC-A

192.168.1.1 → 10.0.12.2 → 10.0.24.2 → 192.168.4.10 (R1, R2, R4, PC-D). The TTL of 125 in the replies from PC-D ($128 - 3$ routers) confirms three router hops.

Q2. Next hop used by R1 for LAN D

10.0.12.2 (R2), as required by rule S4.

2 Part B: RIPv2

All four routers run `router rip` with `version 2` and `no auto-summary`, and advertise their networks with `network 10.0.0.0` and their own LAN. RIP network statements are classful, so 10.0.0.0 covers all of a router's 10.0.x.x/30 links. No static routes are configured.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.13.0/30 is directly connected, GigabitEthernet0/2
L       10.0.13.1/32 is directly connected, GigabitEthernet0/2
R       10.0.23.0/30 [120/1] via 10.0.12.2, 00:00:10, GigabitEthernet0/1
                     [120/1] via 10.0.13.2, 00:00:19, GigabitEthernet0/2
R       10.0.24.0/30 [120/1] via 10.0.12.2, 00:00:10, GigabitEthernet0/1
R       10.0.34.0/30 [120/1] via 10.0.13.2, 00:00:19, GigabitEthernet0/2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.12.2, 00:00:10, GigabitEthernet0/1
R    192.168.3.0/24 [120/1] via 10.0.13.2, 00:00:19, GigabitEthernet0/2
R    192.168.4.0/24 [120/2] via 10.0.12.2, 00:00:10, GigabitEthernet0/1
                    [120/2] via 10.0.13.2, 00:00:19, GigabitEthernet0/2
```

Figure 3: Part B: R1 `show ip route` after convergence. Routes marked R are learned through RIP (administrative distance 120). The route to LAN D (boxed) has hop count 2 and two next hops.

Q3. Hop count and next hops of R1's route to 192.168.4.0/24

Hop count 2, next hops 10.0.12.2 (via R2) and 10.0.13.2 (via R3). Both paths, R1-R2-R4 and R1-R3-R4, are two hops long. RIP's metric ignores link speed, so the paths tie, and IOS installs both and load-balances across them.

Q4. How R1 learns the route without a static route

RIPv2 is a distance-vector protocol. Every router sends its routing table to its neighbours every 30 seconds (RIPv2 uses multicast 224.0.0.9). R4 advertises its directly connected network 192.168.4.0/24 to R2 and R3, which install it with hop count 1 and include it in their own updates to R1. R1 adds one hop to each advertised metric, giving hop count 2 via both R2 (10.0.12.2) and R3 (10.0.13.2). Because the two metrics are equal, both are installed. The route is marked R with administrative distance 120, and it is refreshed by each periodic update, so no manual configuration of the route itself is needed.

3 Part C: OSPF and Link Costs

All routers run OSPF process 1 in area 0, with one `network` statement per directly connected network (e.g. `network 10.0.12.0 0.0.0.3 area 0`, `network 192.168.1.0 0.0.0.255 area 0`). The costs of Table 3 are set with `ip ospf cost` on both ends of every router-to-router link, and LAN interfaces keep their default cost of 1.

Note on cost 1: on a GigabitEthernet interface the default OSPF cost in Packet Tracer is already 1 (100 Mbps reference ÷ 1000 Mbps, rounded up). Because `ip ospf cost 1` equals the default, Packet Tracer does not show it in `show running-config`, even though it was entered. The serial cost of 1 is shown because the serial default is different. Figure 6 confirms that the cost is applied.

3.1 Plan A

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.13.0/30 is directly connected, GigabitEthernet0/2
L       10.0.13.1/32 is directly connected, GigabitEthernet0/2
O       10.0.23.0/30 [110/2] via 10.0.12.2, 00:07:39, GigabitEthernet0/1
O       10.0.24.0/30 [110/2] via 10.0.12.2, 00:07:29, GigabitEthernet0/1
O       10.0.34.0/30 [110/7] via 10.0.12.2, 00:07:02, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
O    192.168.2.0/24 [110/2] via 10.0.12.2, 00:07:39, GigabitEthernet0/1
O    192.168.3.0/24 [110/3] via 10.0.12.2, 00:07:39, GigabitEthernet0/1
O    192.168.4.0/24 [110/3] via 10.0.12.2, 00:07:29, GigabitEthernet0/1
```

Figure 4: Plan A: R1 `show ip route`. Route to LAN D (boxed): `O 192.168.4.0/24 [110/3] via 10.0.12.2`.

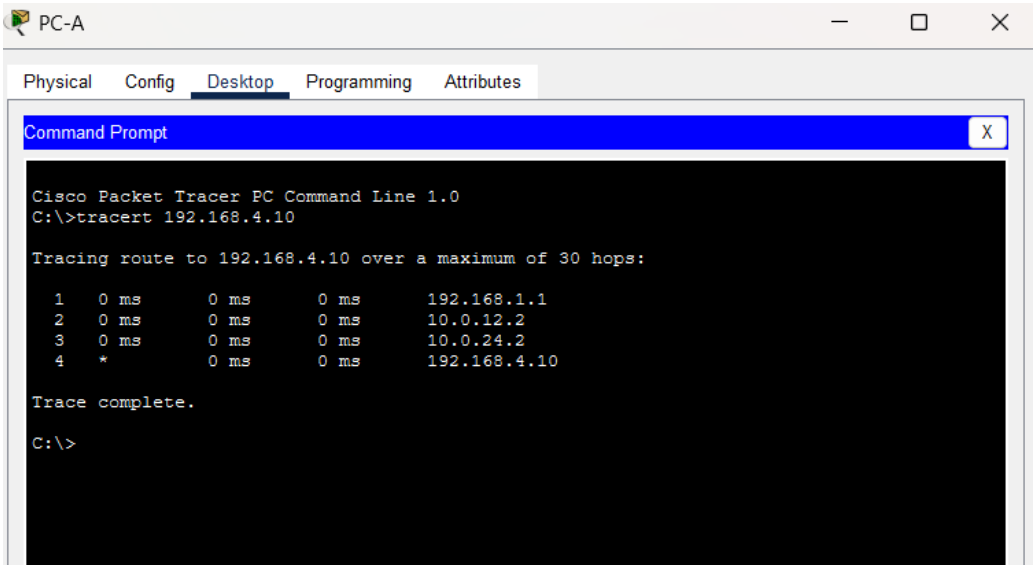


Figure 5: Plan A: `tracert 192.168.4.10` from PC-A follows R1 (192.168.1.1), R2 (10.0.12.2), R4 (10.0.24.2), PC-D.

```
R1#show ip ospf interface GigabitEthernet0/1

GigabitEthernet0/1 is up, line protocol is up
  Internet address is 10.0.12.1/30, Area 0
  Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 192.168.2.1, Interface address 10.0.12.2
  Backup Designated Router (ID) 192.168.1.1, Interface address 10.0.12.1
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.2.1  (Designated Router)
  Suppress hello for 0 neighbor(s)
```

Figure 6: Plan A: `show ip ospf interface GigabitEthernet0/1` on R1 reports `Cost: 1` (boxed), confirming the configured R1–R2 cost.

Q5. Path and metric under Plan A

Path R1-R2-R4, metric 3.

Q6. How the metric is obtained

OSPF adds the cost of each outgoing interface on the way to the destination network: R1 G0/1 (R1→R2) = 1, plus R2 G0/2 (R2→R4) = 1, plus R4 G0/0 (LAN D, default) = 1, which gives 3.

Q7. Why this path is preferred over R1-R3-R4

R1-R3-R4 costs R1 G0/2 (5) + R3 G0/2 (5) + R4 G0/0 (1) = 11. OSPF runs Dijkstra's shortest-path-first algorithm over its link-state database and installs only the lowest-cost path. Since $3 < 11$, R1-R2-R4 is chosen, even though both paths have the same number of hops. For the same reason, only one next hop (10.0.12.2) appears, unlike the two equal next hops RIP showed.

3.2 Plan B

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.13.0/30 is directly connected, GigabitEthernet0/2
L       10.0.13.1/32 is directly connected, GigabitEthernet0/2
O       10.0.23.0/30 [110/6] via 10.0.13.2, 00:11:31, GigabitEthernet0/2
O       10.0.24.0/30 [110/16] via 10.0.13.2, 00:11:06, GigabitEthernet0/2
O       10.0.34.0/30 [110/10] via 10.0.13.2, 00:11:31, GigabitEthernet0/2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
O    192.168.2.0/24 [110/7] via 10.0.13.2, 00:11:31, GigabitEthernet0/2
O    192.168.3.0/24 [110/6] via 10.0.13.2, 00:11:31, GigabitEthernet0/2
O    192.168.4.0/24 [110/11] via 10.0.13.2, 00:11:16, GigabitEthernet0/2
```

Figure 7: Plan B: R1 `show ip route`. Route to LAN D (boxed): `O 192.168.4.0/24 [110/11] via 10.0.13.2`.

```
C:\>tracert 192.168.4.10

Tracing route to 192.168.4.10 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.1
  2    0 ms      0 ms      0 ms      10.0.13.2
  3    0 ms      0 ms      0 ms      10.0.34.2
  4    0 ms      0 ms      0 ms      192.168.4.10

Trace complete.
```

Figure 8: Plan B: `tracert 192.168.4.10` from PC-A follows R1, R3 (10.0.13.2), R4 (10.0.34.2), PC-D.

Q8. Path and metric under Plan B, and why

Path R1-R3-R4, metric 11. Raising the R1–R2 and R2–R4 costs to 10 makes every path that uses them more expensive. The candidate paths (outgoing interface costs, including LAN D's cost of 1) are:

Path	Cost
R1-R2-R4	$10 + 10 + 1 = 21$
R1-R2-R3-R4	$10 + 1 + 5 + 1 = 17$
R1-R3-R2-R4	$5 + 1 + 10 + 1 = 17$
R1-R3-R4	$5 + 5 + 1 = \mathbf{11}$

SPF picks the lowest-cost path, R1-R3-R4. The topology and hop counts did not change, only the costs, and that alone moved the traffic. The same effect is visible for LAN B, which R1 now reaches through R3 and the serial link at cost 7 instead of through its direct R1–R2 link at cost 11.

4 Part D: Link Failure (R2 GigabitEthernet0/2 shut down)

4.1 OSPF (starting from Plan A)

Before the failure (Plan A, from Part C):

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.13.0/30 is directly connected, GigabitEthernet0/2
L       10.0.13.1/32 is directly connected, GigabitEthernet0/2
O       10.0.23.0/30 [110/2] via 10.0.12.2, 00:07:39, GigabitEthernet0/1
O       10.0.24.0/30 [110/2] via 10.0.12.2, 00:07:29, GigabitEthernet0/1
O       10.0.34.0/30 [110/7] via 10.0.12.2, 00:07:02, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
O    192.168.2.0/24 [110/2] via 10.0.12.2, 00:07:39, GigabitEthernet0/1
O    192.168.3.0/24 [110/3] via 10.0.12.2, 00:07:39, GigabitEthernet0/1
O    192.168.4.0/24 [110/3] via 10.0.12.2, 00:07:29, GigabitEthernet0/1
```

Figure 9: OSPF before failure: R1 `show ip route`, LAN D at [110/3] via 10.0.12.2 (boxed).

```
C:\>ping 192.168.4.10

Pinging 192.168.4.10 with 32 bytes of data:

Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 10: OSPF before failure: `ping 192.168.4.10` from PC-A succeeds (TTL 125, three routers).

After the failure (saved as `partD_ospf_linkdown.pkt`):

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.13.0/30 is directly connected, GigabitEthernet0/2
L       10.0.13.1/32 is directly connected, GigabitEthernet0/2
O       10.0.23.0/30 [110/2] via 10.0.12.2, 00:09:04, GigabitEthernet0/1
O       10.0.34.0/30 [110/7] via 10.0.12.2, 00:09:04, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
O    192.168.2.0/24 [110/2] via 10.0.12.2, 00:09:04, GigabitEthernet0/1
O    192.168.3.0/24 [110/3] via 10.0.12.2, 00:09:04, GigabitEthernet0/1
O    192.168.4.0/24 [110/8] via 10.0.12.2, 00:09:04, GigabitEthernet0/1
```

Figure 11: OSPF with R2–R4 down: R1 `show ip route`. 10.0.24.0/30 has disappeared and LAN D is now [110/8] via 10.0.12.2 (boxed).

```
C:\>ping 192.168.4.10

Pinging 192.168.4.10 with 32 bytes of data:

Reply from 192.168.4.10: bytes=32 time=12ms TTL=124
Reply from 192.168.4.10: bytes=32 time=1ms TTL=124
Reply from 192.168.4.10: bytes=32 time=1ms TTL=124
Reply from 192.168.4.10: bytes=32 time=1ms TTL=124

Ping statistics for 192.168.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 12ms, Average = 3ms
```

Figure 12: OSPF with R2–R4 down: `ping 192.168.4.10` from PC-A succeeds (TTL 124, four routers).

```
C:\>tracert 192.168.4.10

Tracing route to 192.168.4.10 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.1
  2    0 ms      0 ms      0 ms      10.0.12.2
  3    4 ms      8 ms      3 ms      10.0.23.2
  4    0 ms      1 ms      0 ms      10.0.34.2
  5    0 ms      0 ms      0 ms      192.168.4.10

Trace complete.

C:\>
```

Figure 13: OSPF with R2–R4 down: `tracert` shows the detour R1, R2 (10.0.12.2), R3 over the serial link (10.0.23.2), R4 (10.0.34.2), PC-D.

After the link is restored (`no shutdown`, state not saved):

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
- BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.13.0/30 is directly connected, GigabitEthernet0/2
L       10.0.13.1/32 is directly connected, GigabitEthernet0/2
O       10.0.23.0/30 [110/2] via 10.0.12.2, 00:20:46,
GigabitEthernet0/1
O       10.0.24.0/30 [110/2] via 10.0.12.2, 00:05:36,
GigabitEthernet0/1
O       10.0.34.0/30 [110/7] via 10.0.12.2, 00:20:46,
GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
O    192.168.2.0/24 [110/2] via 10.0.12.2, 00:20:46,
GigabitEthernet0/1
O    192.168.3.0/24 [110/3] via 10.0.12.2, 00:20:46,
GigabitEthernet0/1
O    192.168.4.0/24 [110/3] via 10.0.12.2, 00:05:36,
GigabitEthernet0/1
```

Figure 14: OSPF after restoring the link: LAN D back at [110/3] via 10.0.12.2 (boxed), and 10.0.24.0/30 is back.

```
C:\>ping 192.168.4.10

Pinging 192.168.4.10 with 32 bytes of data:

Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time=1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 15: OSPF after restoring the link: `ping 192.168.4.10` from PC-A succeeds again with TTL 125.

Q9. OSPF with the link down

Yes, PC-A can still reach PC-D. R1 uses R1-R2-R3-R4 with metric 8 = R1 G0/1 (1) + R2 S0/0/0 (1) + R3 G0/2 (5) + R4 G0/0 (1). The other surviving path, R1-R3-R4, costs $5+5+1 = 11$, so OSPF still goes through R2 and then the cheap serial link to R3. The next hop stays 10.0.12.2, so the change is visible only in the metric ($3 \rightarrow 8$) and in the `tracert`.

Q10. How R1's OSPF table changes after the failure and after restoration

After the failure: R2 and R4 detected the interface going down and flooded updated router LSAs, and every router reran SPF within seconds. In R1's table, 10.0.24.0/30 disappeared (7 subnets became 6), and the LAN D route changed from [110/3] to [110/8]. The next hop stayed 10.0.12.2, but traffic now continues R2 → R3 → R4. Routes that never used the R2–R4 link (LAN B, LAN C, 10.0.23.0/30) were unchanged. 10.0.34.0/30 kept metric 7 but is now reached through R2 and the serial link to R3 instead of through R4.

After restoration: the R2–R4 adjacency re-formed, new LSAs were flooded, and SPF restored the original routes: LAN D at [110/3] via R1-R2-R4, and 10.0.24.0/30 at [110/2]. These two routes have younger ages (00:05:36) than the rest of the table, which shows they were just relearned. Both changes took effect almost immediately, with no hold-down period.

4.2 RIPv2 (starting from Part B)

Before the failure (from Part B):

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.13.0/30 is directly connected, GigabitEthernet0/2
L       10.0.13.1/32 is directly connected, GigabitEthernet0/2
R       10.0.23.0/30 [120/1] via 10.0.12.2, 00:00:10, GigabitEthernet0/1
                     [120/1] via 10.0.13.2, 00:00:19, GigabitEthernet0/2
R       10.0.24.0/30 [120/1] via 10.0.12.2, 00:00:10, GigabitEthernet0/1
R       10.0.34.0/30 [120/1] via 10.0.13.2, 00:00:19, GigabitEthernet0/2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.12.2, 00:00:10, GigabitEthernet0/1
R    192.168.3.0/24 [120/1] via 10.0.13.2, 00:00:19, GigabitEthernet0/2
R    192.168.4.0/24 [120/2] via 10.0.12.2, 00:00:10, GigabitEthernet0/1
                    [120/2] via 10.0.13.2, 00:00:19, GigabitEthernet0/2
```

Figure 16: RIPv2 before failure: R1 `show ip route`, LAN D at [120/2] via 10.0.12.2 and 10.0.13.2 (boxed).

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.4.10

Pinging 192.168.4.10 with 32 bytes of data:

Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 17: RIPv2 before failure: PC-A pings PC-B, PC-C and PC-D (192.168.4.10) successfully.

After the failure, recorded after advancing more than 4 minutes of simulated time (saved as `partD_rip_linkdown.pkt`):

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
- BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E -
EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-
IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.13.0/30 is directly connected, GigabitEthernet0/2
L       10.0.13.1/32 is directly connected, GigabitEthernet0/2
R       10.0.23.0/30 [120/1] via 10.0.12.2, 00:00:25,
GigabitEthernet0/1
                     [120/1] via 10.0.13.2, 00:00:24,
GigabitEthernet0/2
R       10.0.34.0/30 [120/1] via 10.0.13.2, 00:00:24,
GigabitEthernet0/2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.12.2, 00:00:25,
GigabitEthernet0/1
R    192.168.3.0/24 [120/1] via 10.0.13.2, 00:00:24,
GigabitEthernet0/2
R    192.168.4.0/24 [120/2] via 10.0.13.2, 00:00:24,
GigabitEthernet0/2
```

Figure 18: RIPv2 with R2–R4 down: R1 `show ip route`. LAN D is [120/2] via 10.0.13.2 only (boxed), and 10.0.24.0/30 is gone.

```
C:\>ping 192.168.4.10

Pinging 192.168.4.10 with 32 bytes of data:

Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125
Reply from 192.168.4.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.4.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 19: RIPv2 with R2–R4 down: `ping 192.168.4.10` from PC-A succeeds (TTL 125).

```
C:\>tracert 192.168.4.10

Tracing route to 192.168.4.10 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.1
  2    0 ms      0 ms      0 ms      10.0.13.2
  3    0 ms      0 ms      0 ms      10.0.34.2
  4    0 ms      0 ms      0 ms      192.168.4.10

Trace complete.
```

Figure 20: RIPv2 with R2–R4 down: `tracert` follows R1, R3 (10.0.13.2), R4 (10.0.34.2), PC-D.

Q11. RIPv2 with the link down

Yes, connectivity returns. R1's route to LAN D has hop count 2 and a single next hop, 10.0.13.2 (path R1-R3-R4). Before the failure, R1 had two equal 2-hop routes, via R2 and via R3. When the R2–R4 link failed, R2 lost its 1-hop route to LAN D, and the only alternative it still had was through R3, which would cost R1 3 hops. RIP keeps only the routes with the fewest hops, so the R2 next hop was removed and the 2-hop path via R3 remains.

Q12. Comparison of RIPv2 and OSPF in this experiment

Both protocols restored PC-A to PC-D connectivity without any manual route changes, but they behaved differently.

Speed. OSPF routers detected the failure, flooded LSAs and reran SPF, so R1's table changed almost immediately, and the original path returned just as quickly after restoration. RIPv2 learns of changes only through periodic updates (every 30 s), and its invalid and hold-down timers (180 s) delay new information. The assignment therefore required advancing 4 minutes before recording results, and a RIP table can look stable while still waiting to change.

Path chosen. OSPF chose R1-R2-R3-R4 (cost 8) and used the low-cost serial link even though that path has 3 hops. RIPv2 chose R1-R3-R4 because it is the only remaining 2-hop path. The same failure therefore produced different paths, because the protocols measure "best" differently: total configured cost for OSPF, number of hops for RIP.

5 Part E: Comparison (Table 4)

Property	Static routing	RIPv2	OSPF
How are routes obtained?	Typed manually by the administrator with `ip route` on each router (3 per router, 12 in total in Part A).	Learned from neighbours: each router periodically (every 30 s) sends its routing table (distance vector), and the receiver adds 1 hop.	Each router floods LSAs describing its links, builds an identical link-state database, and computes routes with Dijkstra's SPF algorithm.
Adapts automatically to a failed link?	No. Routes stay until the administrator changes them, so with the static routes from Part A, traffic over a failed R2–R4 link would be dropped.	Yes, but slowly. Changes spread through periodic updates and are delayed by invalid/hold-down timers (180 s). In Part D, results were only reliable after waiting 4 minutes.	Yes, quickly. The failure is flooded as LSAs and SPF reruns within seconds (Part D), and the original path returns as fast once the link is restored.
What determines the preferred path?	The administrator's choice of next hop in each route.	Lowest hop count (max 15). Link speed is ignored, and equal-hop paths are load-balanced (two next hops in Part B).	Lowest total cost, the sum of outgoing interface costs (`ip ospf cost` or reference bandwidth ÷ bandwidth). Changing the costs alone moved the path in Plan B.
What must an administrator configure?	Every route to every remote network on every router, and updates whenever the topology changes.	`router rip`, `version 2`, `no auto-summary`, and classful `network` statements on each router.	`router ospf <process>`, `network <address> <wildcard> area <n>` for each network, and optionally `ip ospf cost` to steer traffic.

6 Part F: Reflection

Q13. Two slow links versus three fast links

RIP *would* prefer the route with two slow links, because its only metric is hop count: 2 hops beats 3, whatever the link speeds. OSPF *might* prefer the three fast links, because it chooses the lowest total cost, and by default an interface's cost is reference bandwidth ÷ interface bandwidth. For example, with the default 100 Mbps reference, two 10 Mbps links cost $10 + 10 = 20$, while three 100 Mbps links cost $1 + 1 + 1 = 3$, so OSPF picks the three-link path. It is only "might" because the result depends on the costs: an administrator can set `ip ospf cost` manually, and links faster than the reference bandwidth all get cost 1 unless the reference is raised. Plan B showed this: changing only the costs moved R1's path from R1-R2-R4 to R1-R3-R4, which RIP could never do.

Q14. Manually configuring routes in a network with hundreds of routers

Every router needs a route to every network it is not directly connected to, so the number of static routes grows roughly as routers × networks, which means thousands of entries to type and

maintain. Human error becomes likely: a wrong next hop or a missing route causes black holes or routing loops that are hard to trace. Static routes do not react to failures. When a link goes down, traffic is dropped until an administrator edits routes on many routers, unlike in Part D where RIP and OSPF recovered on their own. Adding a network or changing the topology means updating many routers consistently and in the right order. Paths are also harder to optimise and keep symmetric, and the configuration is hard to audit. Dynamic routing protocols exist to automate exactly this.